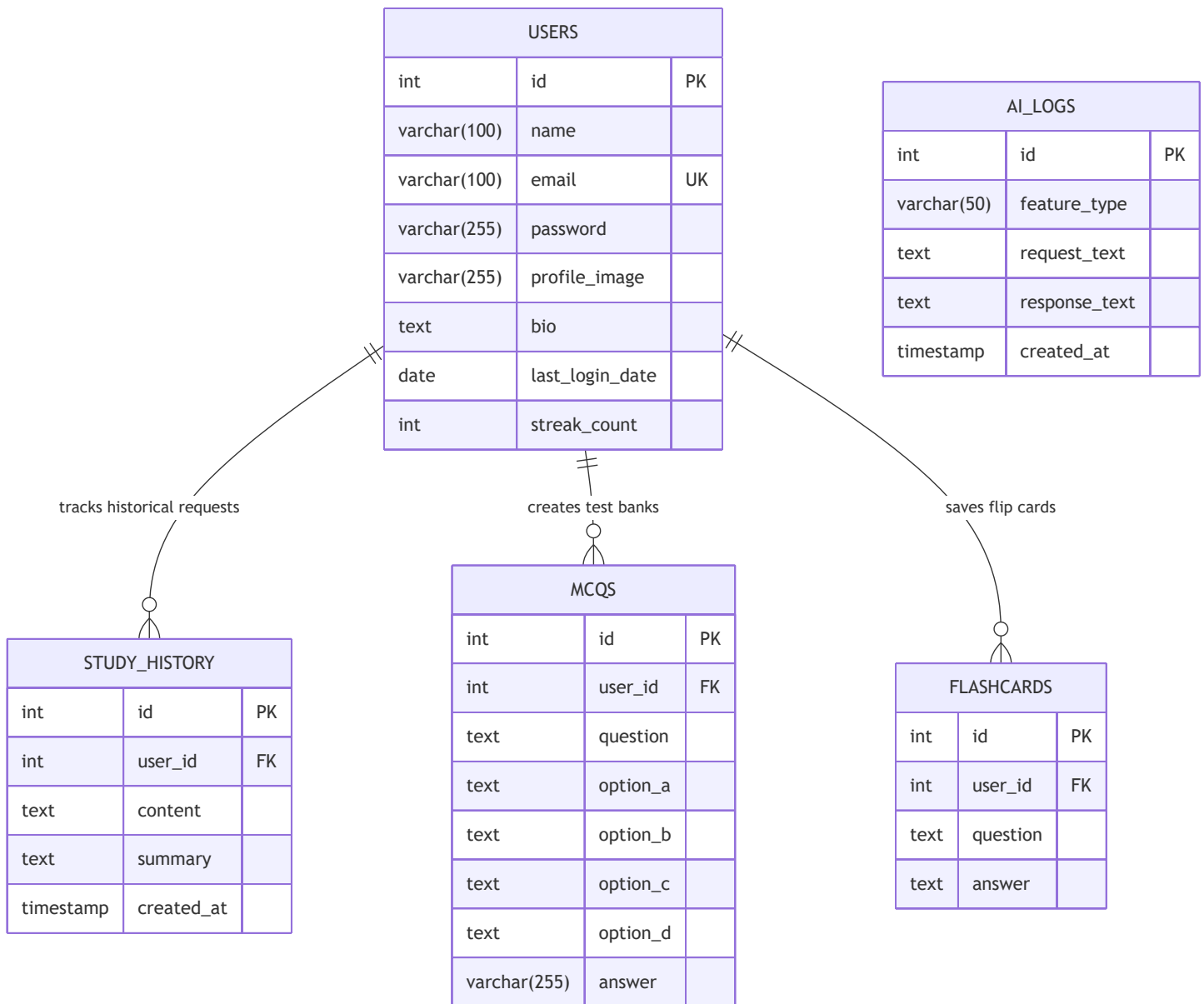


Database Documentation & Mappings

This document outlines the schema design, structural definitions, foreign key integrity constraints, and Object-Relational Mappings (ORM) for the relational MySQL database utilized by the **AI Study Companion** application.

1. Entity-Relationship (ER) Diagram

The diagram below defines the structural tables and relational vectors within the `ai_study_companion` database schema. All domain child tables enforce explicit cascading delete metrics linked directly to parent accounts.



2. Table Schemas & Metadata Descriptions

2.1 users Table

Stores primary user identity records, hashed password credentials, ongoing authentication tracking, and custom profile branding configurations.

Column Name	Data Type	Constraints / Attributes	Default Value	Description
id	INT	Primary Key, Auto	None	Unique primary identifier for

Column Name	Data Type	Constraints / Attributes	Default Value	Description
		Increment		account records.
name	VARCHAR(100)	Nullable	None	Full display name of the registered user.
email	VARCHAR(100)	Unique, Non-Nullable	None	Authentication identity parameter.
password	VARCHAR(255)	Non-Nullable	None	Encrypted/hashed user security token.
profile_image	VARCHAR(255)	Nullable	'default-avatar.png'	Path identifier linking server context uploaded avatars.
bio	TEXT	Nullable	None	Custom markdown-friendly user personal overview.
last_login_date	DATE	Nullable	None	Timetracking snapshot capturing most recent authenticated date.
streak_count	INT	Nullable	0	Active counter summing total continuous daily session days.

2.2 study_history Table

Captures textual user inputs alongside processed structural bullet points outputted by external language generators.

Column Name	Data Type	Constraints / Attributes	Default Value	Description
id	INT	Primary Key, Auto Increment	None	Unique transaction block key.
user_id	INT	Foreign Key	None	References parent users(id) with ON DELETE CASCADE .
content	TEXT	Nullable	None	Original payload text string extracted from forms or file streams.
summary	TEXT	Nullable	None	Formatted markdown response synthesized by Gemini integration.
created_at	TIMESTAMP	Non-Nullable	CURRENT_TIMESTAMP	Static hardware timestamp recording task insertion boundary.

2.3 mcqs Table

Stores generated multi-dimensional multiple choice arrays with verified correct matching vectors.

Column Name	Data Type	Constraints / Attributes	Default Value	Description
id	INT	Primary Key, Auto Increment	None	Unique test item boundary key.
user_id	INT	Foreign Key	None	References parent users(id) with ON DELETE CASCADE .

Column Name	Data Type	Constraints / Attributes	Default Value	Description
question	TEXT	Nullable	<i>None</i>	Interrogative question prompt extracted from subject domains.
option_a	TEXT	Nullable	<i>None</i>	Distractor choice variable parameter.
option_b	TEXT	Nullable	<i>None</i>	Distractor choice variable parameter.
option_c	TEXT	Nullable	<i>None</i>	Distractor choice variable parameter.
option_d	TEXT	Nullable	<i>None</i>	Distractor choice variable parameter.
answer	VARCHAR(255)	Nullable	<i>None</i>	Validated correct answer selection matching distractor identifiers.

2.4 flashcards Table

Maintains reactive review parameters focused on two-way active recall study logic.

Column Name	Data Type	Constraints / Attributes	Default Value	Description
id	INT	Primary Key, Auto Increment	<i>None</i>	Unique item tracking code.
user_id	INT	Foreign Key	<i>None</i>	References parent <code>users(id)</code> with <code>ON DELETE CASCADE</code> .
question	TEXT	Nullable	<i>None</i>	Concise front-face concept target string.
answer	TEXT	Nullable	<i>None</i>	Detailed back-face explanation text payload.

2.5 ai_logs Table

Decoupled traceability table engineered to safely capture raw JSON payloads and API telemetry independent of application domain constraints.

Column Name	Data Type	Constraints / Attributes	Default Value	Description
id	INT	Primary Key, Auto Increment	None	Global diagnostics record pointer.
feature_type	VARCHAR(50)	Nullable	None	Feature routing tag (e.g., 'Summary' , 'MCQ' , 'Flashcards').
request_text	TEXT	Nullable	None	Complete outward context string payload dispatched to the model.
response_text	TEXT	Nullable	None	Complete raw JSON or streaming output returned by upstream APIs.
created_at	TIMESTAMP	Non-Nullable	CURRENT_TIMESTAMP	Explicit execution entry snapshot.

3. Object-to-Table Class Mappings

The table below correlates backend models, executing Data Access Objects (DAOs), and related downstream consumer components to specific relational targets.

Relational Table Target	Data Transfer Object (POJO)	Orchestrating Data Access Object	Primary Controllers / Servlets Utilizing Flow
users	com.aistudy.model.User	com.aistudy.dao.UserDAO	LoginServlet , RegisterServlet , ProfileServlet
study_history	<i>Decoupled / Primitive Map List</i>	com.aistudy.dao.StudyDAO	SummaryServlet , DashboardServlet
mcqs	com.aistudy.model.MCQ	com.aistudy.dao.StudyDAO	MCQServlet , DashboardServlet
flashcards	com.aistudy.model.Flashcard	com.aistudy.dao.StudyDAO	FlashcardServlet , DashboardServlet
ai_logs	<i>Stateless Infrastructure Log</i>	com.aistudy.dao.AILogDAO	SummaryServlet , MCQServlet , FlashcardServlet